

ECS200 Workshop Report

Summing and Difference Amplifiers

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Abstract—This experiment evaluates the operational behavior of summing and difference amplifier circuits constructed using the LM741 operational amplifier and the NI myDAQ platform. In the first exercise, a weighted summing amplifier was designed and tested on three distinct combinations of input voltages and resistor values (R_f , R_1 , R_2). The measured DC output voltages were compared with theoretical calculations, yielding percentage errors ranging from 0.95% to 8.77%. In the second exercise, a difference amplifier was configured with two AC input signals ($V_1 = +5V_p$ and $V_2 = -5V_p$). For a calculated input difference of 10.000V, the measured output voltage was 10.049V (20.098V_{pp}), demonstrating an experimental circuit gain of 1.005. The findings validate the theoretical mathematical operations of summing and difference op-amp configurations within low error margins.

I. INTRODUCTION

Operational amplifiers are the foundational components in modern analog circuit design used for signal conditioning and arithmetic computations[3]. This report presents the implementation and performance evaluation of the summing and difference amplifier circuits constructed using the LM741 op-amp and the NI myDAQ platform[1], [3].

A. Background and Objectives

Operational amplifiers (op-amps) are fundamental building blocks in analog electronics, widely utilized for signal processing, filtering, and performing mathematical operations[3]. The LM741 is an eight-pin operational amplifier IC commonly used to construct summing and difference amplifiers [3]. Pin 6 serves as the output, while pins 2 and 3 serve as the inverting and non-inverting inputs, respectively[3]. Pin 8 remains unused, and supply voltages are applied to pins 4 and 7 [3]. For the summing amplifier configuration, the output voltage is governed by the relation[1]:

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right) \quad (1)$$

where R_f represents the feedback resistor, R_1 and R_2 represent the input resistors, and V_1 and V_2 represent the input voltages as shown in[1].

B. Equipment Used

The hardware components and tools utilized during this experiment, as shown in Figure 1:

- NI myDAQ
- myDigital Protoboard
- 741 Operational Amplifier IC (LM741)
- Resistors: 3.3Ω, 4.7Ω, 10Ω, 22kΩ, 47Ω, 100Ω, and 330Ω.
- Multimeter leads and jumper wires.

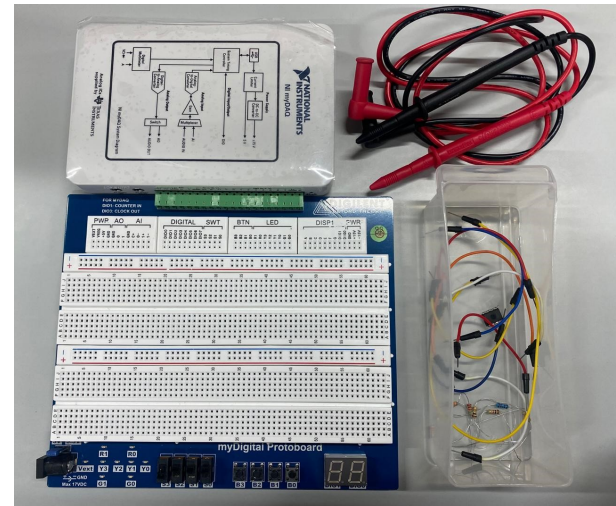


Fig. 1. Equipment Used

II. DESIGN

A. Exercise 1—Summing Amplifier

The weighted summing amplifier circuit was designed by placing the LM741 operational amplifier chip on the myDigital Protoboard[2]. Operational supply voltages were provided by connecting Pin 7 to +15V and Pin 4 to -15V, while Pin 3 (non-inverting input) was tied directly to ground as shown in[3]. A feedback resistor (R_f) was connected between Pin 2 (inverting input) and Pin 6 (output)[1]. Input resistor R_1 was wired from Pin 2 to a fixed +5V DC source (V_1), and R_2 was wired from Pin 2 to analog output AO-0 to supply V_2 ranging from -10V to +10V [1]. Channel AO-0 was jumpered to AI-0+ to monitor input and output waveforms concurrently[1]. Output DC voltages at Pin 6 were measured using the NI myDAQ digital voltmeter with the red lead connected to "V" and the black lead to "COM", as shown in Figure 2

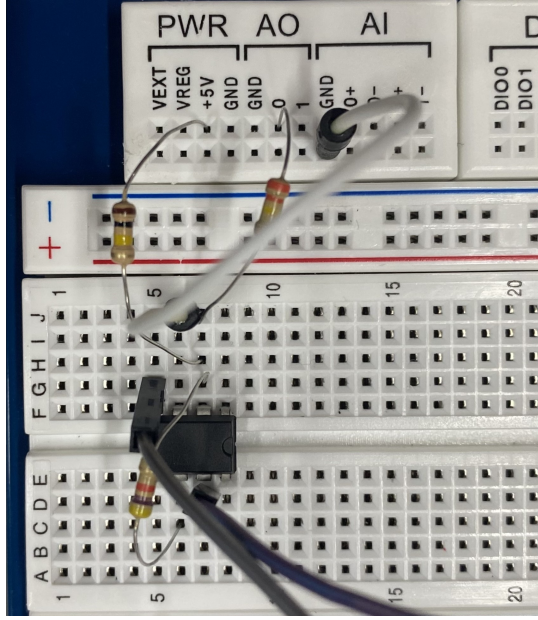


Fig. 2. Summing amplifier circuit connections. .

B. Exercise 2—Difference Amplifier

The difference amplifier circuit was constructed on the myDigital Protoboard to analyze the output waveform, peak-to-peak voltage (V_{pp}), and circuit gain[2]. Operational supply voltages of $+15V$ and $-15V$ were supplied to Pin 7 and Pin 4 of the LM741 IC, respectively[3]. The circuit was driven using two AC input signals configured as $V_1 = +5V_p$ and $V_2 = -5V_p$, yielding a calculated input differential voltage ($V_2 - V_1$) of $10V$ [1]. Pin 6 (output) was connected to the AI-0+ channel of the NI myDAQ to observe the output waveform concurrently with the input on the virtual oscilloscope, as shown in[1]. The measured peak-to-peak output voltage (V_{pp}) was recorded and compared against the calculated input differential voltage to evaluate circuit gain and operational response[1].

III. EXPERIMENTAL TESTING AND RESULTS

The constructed operational amplifier circuits were tested experimentally using the NI myDAQ virtual instruments to evaluate performance against theoretical calculations.

A. Exercise 1—Summing Amplifier

The summing amplifier was evaluated across six resistor and input voltage (V_2) combinations by comparing theoretical values from Equation 1 with DC voltages measured via the NI myDAQ digital voltmeter, as shown in Table I and Figure 3.

TABLE I
SUMMING AMPLIFIER EXPERIMENTAL RESULTS

R_f (k Ω)	R_1 (k Ω)	R_2 (k Ω)	V_1 (V)	V_2 (V)	Calculated V_{out} (V)	Measured V_{out} (V)	% Error
47	100	330	+5	+2	-2.635	-2.610	0.95
22	22	47	+5	-10	-0.319	-0.291	8.77
3.3	10	4.7	+5	+5	-5.161	-5.013	2.86

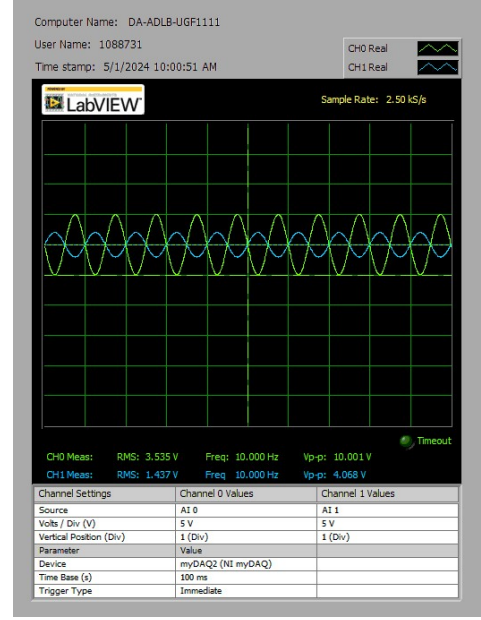


Fig. 3. Output signal ($2 V_p$) displayed with the input signal ($5 V_p$) on the oscilloscope

B. Exercise 2—Difference Amplifier

The difference amplifier was tested using two complementary AC input signals ($V_1 = +5V_p$, $V_2 = -5V_p$) to record input differential voltage, peak-to-peak output voltage (V_{pp}), and differential gain on the NI myDAQ virtual oscilloscope, as shown in Table II

TABLE II
DIFFERENCE AMPLIFIER EXPERIMENTAL RESULTS

Calculated $V_2 - V_1$ (V)	Measured V_{pp} (V)	Measured $ V_{out} $ (V)	Gain ($V_{out}/(V_2 - V_1)$)
10.000	20.098	10.049	1.005

IV. CONCLUSION

The oscilloscope measurements confirmed that the difference amplifier effectively subtracted the two AC signals while maintaining unified gain. The experimental output amplitude of $10.049V$ ($20.098V_{pp}$) demonstrated an exceptionally low error margin of 0.49% compared to the theoretical $10.000V$ prediction. The output waveform exhibited clean sinusoidal symmetry without noticeable clipping or phase distortion, verifying correct IC biasing and proper common-mode rejection performance[1], [3].

REFERENCES

- [1] USER GUIDE AND SPECIFICATIONS NI myDAQ. (n.d.). National Instruments Corporation.
- [2] myDigital™ Board Reference Manual. (2014, November 14). Mydigital-Rm.
- [3] Subedi, B. (2023). OP amp IC 741 basics, characteristics, pins, applications. How To Electronics. Available at: <https://how2electronics.com/op-amp-ic-741-basics-characteristics-pins-applications/>